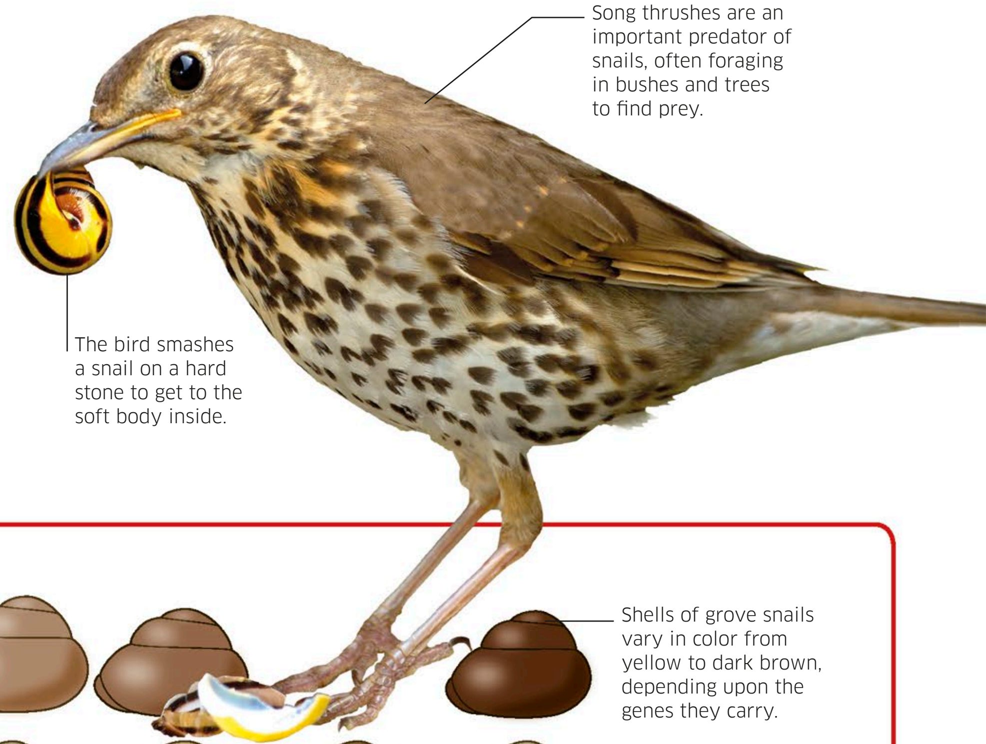
Evolution

All living things are related and united by a process called evolution. Over millions of years, evolution has produced all the species that have ever lived.

Change is a fact of life. Every organism goes through a transformation as it develops and gets older. But over much longer periods of time—millions or billions of years—entire populations of plants, animals, and microbes also change by evolving. All the kinds of organisms alive today have descended from different ones that lived in the past, as tiny variations throughout history have combined to produce entirely new species.



Natural selection

The characteristics of living things are determined by genes (see pp.180–181), which sometimes change as they are passed down through generations—producing mutations. All the variety in the natural world—such as the colors of snail shells—comes from chance mutations, but not all of the resulting organisms do well in their environments. Only some survive to pass their attributes on to future generations—winning the struggle of natural selection.

Dry grassy habitat
Against a background of dry grass, snails with darker shells are most easily spotted, causing the paler ones to survive in greater numbers.

Dark woodland habitat
In woodland, grove snails with shells that match the dark brown leaf litter of the woodland floor are camouflaged and survive, but yellow-shelled snails are spotted by birds.

Hedge habitat
In some sun-dappled habitats with a mixture of grass, twigs, and leaves, stripy-shelled grove snails are better disguised, and plain brown or yellow ones become prey.